

PROBABILITY THEORY

MTM2592-Gr:2

SECOND WEEK

Assc. Prof. Dr. Ulku (BABUSCU) YESIL

ubabuscu@yildiz.edu.tr

ulkubabuscu@gmail.com

<https://avesis.yildiz.edu.tr/ubabuscu>

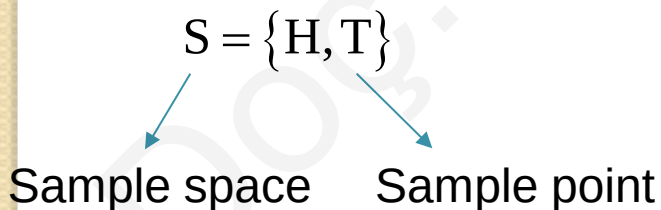
Probability Concepts

EXPERIMENT:

Statisticians use the word 'experiment' to describe any process that generates a set of data. For example; tossing a coin is an experiment. It generates 'Head' and 'Tail'. If an experiment always generates the same outcome under the same conditions, it is called a **deterministic experiment**. Otherwise, it is called a **random experiment**. In this course we deal with random experiments.

SAMPLE POINT & SAMPLE SPACE:

The set of all possible outcomes of an experiment is called sample space, and it is usually denoted by the letter S. Each outcome of the experiment is called a sample point of S. Sample points build n-dimensional sample space.



$$\begin{aligned} S &= \{(x, y) \mid x = 1, 2, \dots, 6; y = 1, 2, \dots, 6\} \\ &= \{(1, 1), (1, 2), \dots, (6, 6)\} \end{aligned}$$

↓
Sample point

DISCRETE AND CONTINUOUS SAMPLE SPACE:

A sample space that contains a finite number of elements or countable infinite elements is called a discrete sample space. Otherwise, it is called a continuous sample space.

For example; $S = \{1, 2, 3, 4, 5, 6\}$ $\longrightarrow$ discrete sample space

$S = \{x | 2 < x < 5\}$ $\longrightarrow$ continuous sample space

EVENT:

An event is a subset of a sample space. A sample point or several sample points are called an **event**. An event is a subset of a sample space. If an event corresponds to a single point of a sample space, it is said to be a **simple (elementary) event**. If it corresponds to more than one point, then it is called a **compound event**.

Sample spaces and events, particularly relationships among events, are often depicted by means of **Venn diagrams**, in which the sample space is represented by a rectangle, while events are represented by regions within the rectangle, usually by circles or parts of circles. For instance, the shaded regions of the four Venn diagrams of Figure 3 represent, respectively, event A , the complement of event A , the union of events A and B , and the intersection of events A and B . When we are dealing with three events, we usually draw the circles as in Figure 4. Here, the regions are numbered 1 through 8 for easy reference.

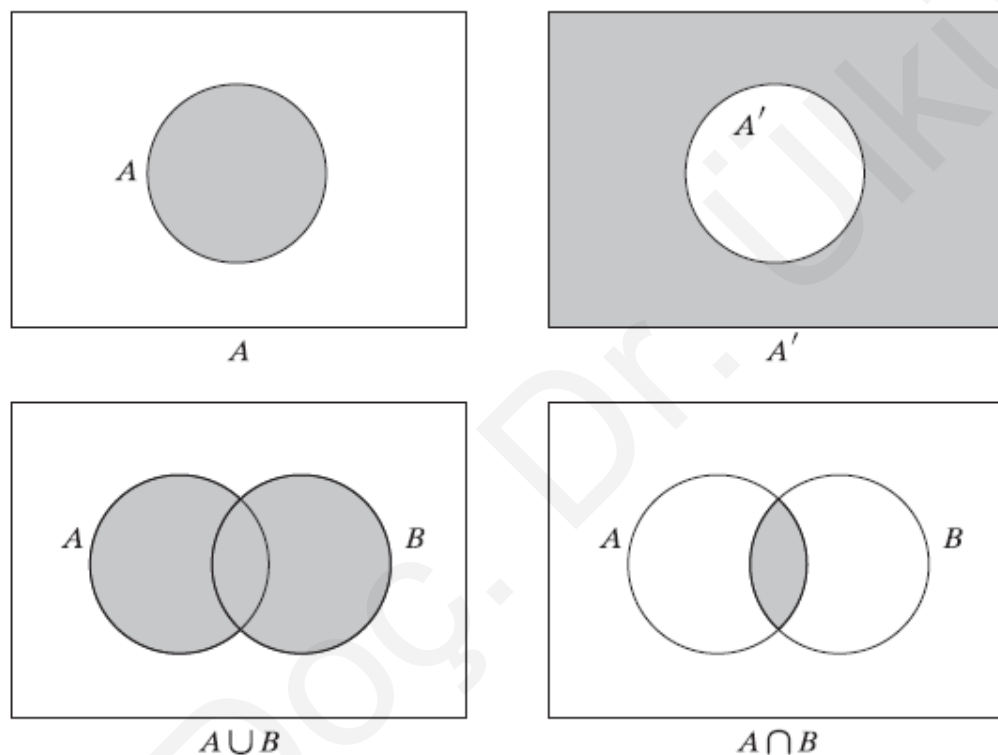


Figure 3. Venn diagrams.

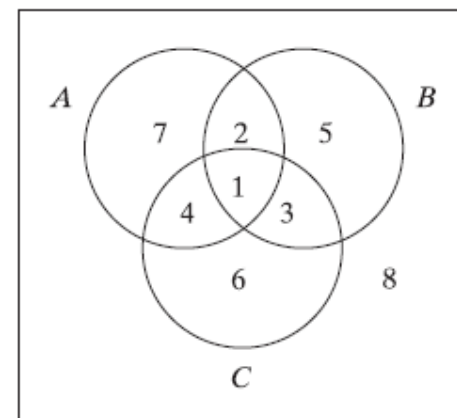
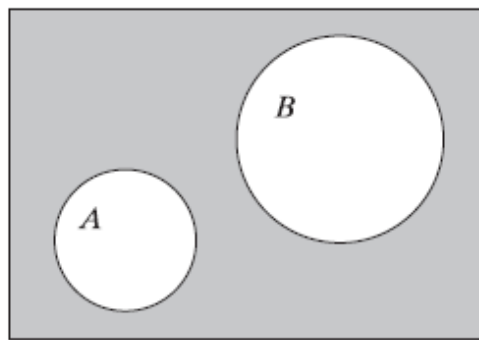
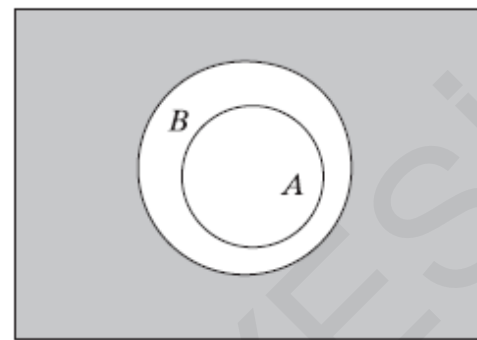


Figure 4. Venn diagram.



A and B mutually exclusive



A contained in B

Figure 5. Diagrams showing special relationships among events.

To indicate special relationships among events, we sometimes draw diagrams like those of Figure 5. Here, the one on the left serves to indicate that events *A* and *B* are **mutually exclusive**.

DEFINITION 3. MUTUALLY EXCLUSIVE EVENTS. *Two events having no elements in common are said to be mutually exclusive.*

When *A* and *B* are mutually exclusive, we write $A \cap B = \emptyset$, where $\emptyset$ denotes the **empty set**, which has no elements at all. The diagram on the right serves to indicate that *A* is contained in *B*, and symbolically we express this by writing $A \subset B$.

Example: If a random experiment consist of tossing a balanced die, then the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A_1 = \{1, 3, 5\} \quad \rightarrow \quad \text{The event of getting odd numbers}$$

$$A_2 = \{2, 4, 6\} \quad \rightarrow \quad \text{The event of getting even numbers}$$

$$A_3 = \{1, 2\} \quad \rightarrow \quad \text{The event of getting a number smaller than 3}$$

Example: Lifetime of produced bulbs

$$S = \{x \mid 0 \leq x < \infty\}$$

$$B_1 = \{x \mid x > 200\} \quad \rightarrow \quad \text{The event of lasting longer than 200 hours}$$

Example: Let a balanced coin be tossed three times. Construct a sample space and an event that the coin shows head at least two times.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$A = \{HHH, HHT, HTH, THH\}$$

Example: Let $S = \{x | 1 \leq x \leq 5\}$ be a sample space and

$A = \{x | 1 \leq x \leq 4\}$, $B = \{x | 2 < x \leq 5\}$ be two events. Construct the events

$$A \cup B = \{x | 1 \leq x \leq 5\}$$

$$A \cup \bar{B} = \{x | 1 \leq x \leq 4\}$$

$$\bar{A} \cap B = \{x | 4 < x \leq 5\}$$

$$\overline{A \cap B} = \{x | 1 \leq x \leq 2 \text{ or } 4 < x \leq 5\}$$

THE DEFINITIONS OF THE PROBABILITY:

Conventional Definition: Let S be a sample space that contains equally likely outcomes and A be an event defined on S . The probability of the event A is

$$P(A) = \frac{n(A)}{n(S)}$$

Ex: What is the probability that a balanced die comes up greater than 4?

$$S = \{1, 2, 3, 4, 5, 6\} \quad A = \{5, 6\} \quad P(A) = \frac{n(A)}{n(S)} = \frac{2}{6}$$

Frequency Definition: If N is the number of trials of an experiment and n is the number of times the event A happens, then

$$p(A) = \lim_{N \rightarrow \infty} \frac{n}{N}$$

Subjective Definition: The use of intuition, personal beliefs and other indirect information in arriving at probabilities is referred to as the subjective definition of probability.

x : The chance of occurrence of an event A

y : The chance of non-occurrence of an event A

$$\frac{p(A)}{1 - p(A)} = \frac{x}{y} \Rightarrow p(A) = \frac{x}{x + y} \quad (\text{subjective chance})$$

Ex: An expert considers that the chance of Besiktas's winning the cup is 2:5. According to this expert what is the probability that Besiktas will win the cup?

$$P = 2/(2+5) = 2/7$$

PROBABILITY AXIOMS:

Definition: A real valued function P , defined on a sample space S is called a probability function if the following axioms are satisfied.

1) $0 \leq P(A) \leq 1$ for all events A

2) $P(S)=1$

3) If $A_1, A_2, \dots, A_n$ are disjoint events (mutually exclusive), then

$$P(A_1 \cup A_2 \cup A_3 \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

For two disjoint events



$$P(A \cup B) = P(A) + P(B)$$

Theorem 1: $P(\emptyset) = 0$ where $\emptyset$ is an empty set.

Proof: A is any set.

$$A \cap \emptyset = \emptyset \quad \text{ve} \quad A \cup \emptyset = A$$

$$P(A) = P(A \cup \emptyset) = P(A) + P(\emptyset) \Rightarrow P(\emptyset) = 0$$

Theorem 2: $P(\bar{A}) = 1 - P(A)$ where $\bar{A}$ is the complement of A.

Proof: $S = A \cup \bar{A}$ and $A \cap \bar{A} = \emptyset$

$$1 = P(S) = P(A \cup \bar{A}) = P(A) + P(\bar{A})$$

$$P(\bar{A}) = 1 - P(A)$$

Theorem 3: If $A \subset B$ then $P(A) \leq P(B)$

Proof: $(B \setminus A) \cap A = \emptyset$, $(B \setminus A) \cup A = B$

$$P((B \setminus A) \cup A) = P(B \setminus A) + P(A) = P(B)$$

$$P(B \setminus A) \geq 0 \Rightarrow P(B) \geq P(A)$$

Remark: $A \subset S$ then $P(A) \leq P(S)$

$$P(S) = 1, P(A) \geq 0$$

$$\text{then } 0 \leq P(A) \leq P(S) \Rightarrow 0 \leq P(A) \leq 1$$

Theorem 4:

$$P(A/B) = P(A) - P(A \cap B)$$

Proof: $A = (A/B) \cup (A \cap B)$ since $(A/B) \cap (A \cap B) = \emptyset$

$$P(A) = P((A/B) \cup (A \cap B)) = P(A/B) + P(A \cap B)$$

$$P(A/B) = P(A) - P(A \cap B)$$

Theorem 5: If A and B are any event then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: $A \cup B = (A/B) \cup B$

By using Theorem 4

$$\begin{aligned} P(A \cup B) &= P(A/B) + P(B) = P(A) - P(A \cap B) + P(B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

THEOREM If A , B , and C are any three events in a sample space S , then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\ - P(B \cap C) + P(A \cap B \cap C)$$

Proof

$$\begin{aligned} P(A \cup B \cup C) &= P[A \cup (B \cup C)] \\ &= P(A) + P(B \cup C) - P[A \cap (B \cup C)] \\ &= P(A) + P(B) + P(C) - P(B \cap C) \\ &\quad - P[A \cap (B \cup C)] \end{aligned}$$



$$\begin{aligned} P[A \cap (B \cup C)] &= P[(A \cap B) \cup (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P[(A \cap B) \cap (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \end{aligned}$$

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) \\ &\quad - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

Theorem 6: If $A_1, A_2, \dots, A_n$ are any event then;

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n-1} P(A_1 \cap A_2 \dots \cap A_n)$$

BOOLE INEQUALITY: $\Rightarrow P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$

If $A_1, A_2, \dots, A_n$ are any event $\Rightarrow$

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

BONFERONNI INEQUALITY $\Rightarrow$

$$P(A_1 \cap A_2) \geq P(A_1) + P(A_2) - 1$$

General form of this inequality $\Rightarrow$

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - 1$$

Bonferroni's inequality is used to obtain a lower limit where it is impossible to calculate the probability of intersection.

Example: If $P(A)=0.90$, $P(B)=0.95$ create a lower limit for $P(A \cap B)$

Solution: $P(A \cap B) \geq P(A) + P(B) - 1 = 0.90 + 0.95 - 1 = 0.85$

Theorem 7: If A and B are any events, than

$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

Example: A card is randomly selected from a deck of 52 cards. Let;

$A = \{\text{The card is Spades}\}$

$B = \{\text{The card is colorful(Jack, Queen or King)}\}$

Find $P(A)$, $P(B)$ and
 $P(A \cap B)$

Note:

Jack $\rightarrow$ Bacak

Queen $\rightarrow$ Kız

King $\rightarrow$ Papaz

Ace $\rightarrow$ As

Hearts $\rightarrow$ Kupa

Spades $\rightarrow$ Maça

Diamonds $\rightarrow$ Karo

Clubs $\rightarrow$ Sinek

Solution:

Example: Two items are randomly chosen from a set of 12 items of which 4 are defective.

$A = \{\text{Both of the chosen items are defective}\}$

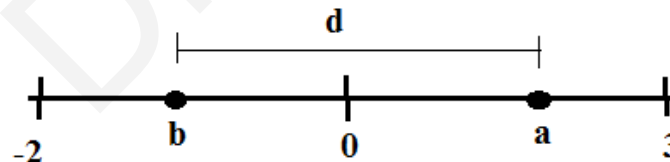
$B = \{\text{Neither of the chosen items are defective}\}$

$C = \{\text{At least one of the chosen items are defective}\}$

Find $P(A)$, $P(B)$ and $P(C)$.

Solution:

Example: Points a and b are randomly selected on a $\mathbb{R}$ real line as shown below where $-2 \leq b \leq 0$ and $0 \leq a \leq 3$. What is the probability that the distance d between a and b is longer than 3?



Solution:

Example: The probability that a student passes from mathematics is $\frac{4}{5}$, the probability that a student passes from physics is $\frac{2}{3}$ and the probability that the student passes from both subjects is $\frac{1}{2}$. Find the probability that the student

- a) Fails physics,
- b) Passes mathematics or physics,
- c) Passes mathematics or fails physics.

Solution:

Example: If two dies are rolled, what is the probability that the total number comes up 9?

Solution:

Example: A coin is tossed three times.

$A = \{\text{Comes up head at least two times}\}$

$B = \{\text{Comes up head at the second tossing}\}$

$C = \{\text{Comes up head only at the second tossing}\}$

- a) Are A and B disjoint events,
- b) Are A and C disjoint events,
- c) Are B and C disjoint events,
- d) Find $P(A \cup B)$, $P(A \cup C)$, $P(B \cup C)$

CONDITIONAL PROBABILITY:

Let A and B are two events defined on a sample space S. Then, the probability of the event A occurring when it is known that the event B has occurred is called a conditional probability and it is denoted by

$$P(A|B)$$

The symbol $P(A|B)$ is usually read «the probability that A occurs given that B occurs» or simply «the probability of A, given B». We can consider

$$P(A) = P(A|S)$$

Thus, B is said to be a reduced sample space for this situation.

Definition: The conditional probability of A, given B denoted by $P(A|B)$ is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Example: The people in a small town are categorized as shown in the table:

	Employed (E)	Unemployed (U)
Male (M)	300	400
Female (F)	250	250

Find $P(U|M)$, $P(M|U) = ?$

Example: It is known that the outcomes is an even number when a balanced die is tossed. Accordingly, what is the probability that this number is 4?

Example: 100 people are surveyed in order to determine relationship between high tension and obesity. The results are given as:

	Thin	Normal	Fat	Total
High Tension	8	10	22	40
Non-high Tension	24	16	20	60
Total	32	26	42	100

Find the probability that a randomly selected person

- a) is fat,
- b) has high tension,
- c) is fat given that he/she has high tension
- d) has high tension given that he/she is fat.

Example: Let two dies be rolled. What is the probability that both of them come up 4 given that they come up 8 in total.

Solution:

Example: Two balls successively selected without replacing them back, from a bag that contains 3 green balls and 5 red balls. What is the probability that the second chosen ball is green given that the first ball is green?

Solution:

MULTIPLICATION RULE FOR CONDITIONAL PROBABILITY:

* The probability of occurrence of the intersection of two events can be calculated using the product rule.

$$P(A_1|A_2) = \frac{P(A_1 \cap A_2)}{P(A_2)} \Rightarrow P(A_1 \cap A_2) = P(A_1|A_2).P(A_2) \quad P(A_2) > 0$$

or

$$P(A_2|A_1) = \frac{P(A_1 \cap A_2)}{P(A_1)} \Rightarrow P(A_1 \cap A_2) = P(A_2|A_1).P(A_1) \quad P(A_1) > 0$$

*In the case of three events;

$$P(A_3|A_1 \cap A_2) = \frac{P(A_1 \cap A_2 \cap A_3)}{P(A_1 \cap A_2)}$$

$$P(A_1 \cap A_2 \cap A_3) = \underbrace{P(A_1 \cap A_2)} P(A_3|A_1 \cap A_2)$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$

*If this formula is generalized for n events $A_1, A_2, \dots, A_n$,

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1).P(A_2|A_1).P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1).P(A_2|A_1).P(A_3|A_1 \cap A_2) \dots P\left(A_n \middle| \left(\bigcap_{i=1}^{n-1} A_i\right)\right)$$

Example: 4 bulbs are defective in a box of 18 bulbs. 2 bulbs are successively selected without replacing them back. What is the probability that

- a) Both bulbs are defective,
- b) Both bulbs are non-defective,
- c) The first chosen bulb is defective and the second one is non-defective,
- d) The first chosen bulb is non-defective and the other one is defective,

Solution:

Example: 4 bulbs are defective in a box of 15 bulbs. 3 bulbs are successively selected without replacing them back. What is the probability that all bulbs are non-defective?

Solution:

Example: A box contains 8 blue balls and 6 red balls. 3 balls are selected successively from the box. Find the probability that the first ball is red, the second ball is blue and third one is red, if 3 balls are chosen

- a) Without replacing it back,
- b) Replacing them back.

Solution:

DISCRETIZATION OF SAMPLE SPACE:

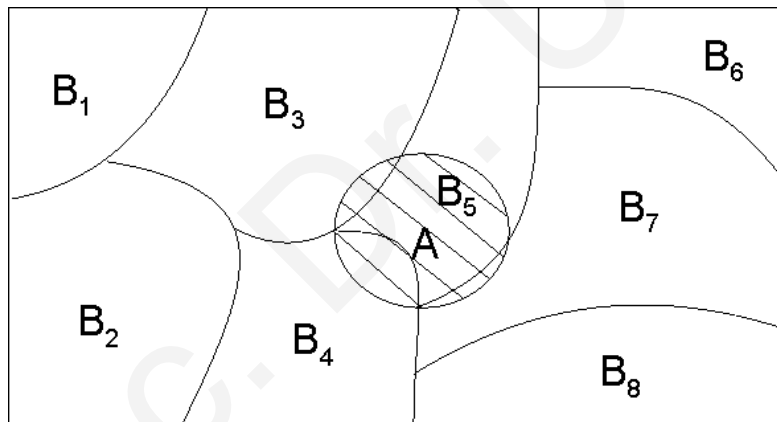
Let $B_i \subset S$,

$$a) \quad P(B_i \cap B_j) = \emptyset \quad i \neq j$$

$$b) \quad \bigcup_{i=1}^n B_i = S$$

$$c) \quad P(B_i) > 0 \quad \text{for all } i$$

Then $B_1, B_2, \dots, B_n$ events show a discretization of the sample space S .
The figure shows discretization of S for $n = 8$.



Theorem: According to the definition given above,

$$P(B_1) + P(B_2) + \dots + P(B_n) = 1$$

Theorem: If B_i ($i=1,2,\dots,n$) is a discretization of a sample space S , then

$$P(A) = \sum_{i=1}^n P(B_i \cap A) = \sum_{i=1}^n P(B_i) \cdot P(A|B_i)$$

for any events A in S .

Proof:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$

$$P(A \cap B_i) = P(B_i)P(A|B_i)$$

$$P(A) = P(B_1)P(A|B_1) + \dots + P(B_n)P(A|B_n)$$

Since $A \cap B_i$
are disjoint events

Example: A type of machine is produced in the factories A, B and C. The number of machines produced in A is 2 times of machines produced in B and also in C. It is known that 2% of machines produced in A as well as in B, and 4% of machines produced in C are defective. If a machine is randomly selected from the store in which all produced machines put, what is the probability that the machine is defective?

Solution: